

copy-of-da-project

February 5, 2024

```
[ ]: import pandas as pd

from google.colab import files
uploaded = files.upload()

df = pd.read_csv('AB_NYC_2019.csv')

print("Original dataset info:")
print(df.info())

# Handling missing values
df.dropna(inplace=True)

# Removing duplicates
df.drop_duplicates(inplace=True)

# Display cleaned dataset info
print("\nCleaned dataset info:")
print(df.info())

# Save cleaned dataset to a new CSV file
df.to_csv('DA_airbnb_data_cleaned.csv', index=False)
```

<IPython.core.display.HTML object>

Saving AB_NYC_2019.csv to AB_NYC_2019 (1).csv

Original dataset info:

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 48895 entries, 0 to 48894

Data columns (total 16 columns):

#	Column	Non-Null Count	Dtype
0	id	48895 non-null	int64
1	name	48879 non-null	object
2	host_id	48895 non-null	int64
3	host_name	48874 non-null	object
4	neighbourhood_group	48895 non-null	object
5	neighbourhood	48895 non-null	object

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6   latitude          48895 non-null float64
7   longitude         48895 non-null float64
8   room_type         48895 non-null object
9   price             48895 non-null int64
10  minimum_nights    48895 non-null int64
11  number_of_reviews 48895 non-null int64
12  last_review       38843 non-null object
13  reviews_per_month 38843 non-null float64
14  calculated_host_listings_count 48895 non-null int64
15  availability_365   48895 non-null int64

```

dtypes: float64(3), int64(7), object(6)

memory usage: 6.0+ MB

None

Cleaned dataset info:

<class 'pandas.core.frame.DataFrame'>

Int64Index: 38821 entries, 0 to 48852

Data columns (total 16 columns):

#	Column	Non-Null Count	Dtype
0	id	38821 non-null	int64
1	name	38821 non-null	object
2	host_id	38821 non-null	int64
3	host_name	38821 non-null	object
4	neighbourhood_group	38821 non-null	object
5	neighbourhood	38821 non-null	object
6	latitude	38821 non-null	float64
7	longitude	38821 non-null	float64
8	room_type	38821 non-null	object
9	price	38821 non-null	int64
10	minimum_nights	38821 non-null	int64
11	number_of_reviews	38821 non-null	int64
12	last_review	38821 non-null	object
13	reviews_per_month	38821 non-null	float64
14	calculated_host_listings_count	38821 non-null	int64
15	availability_365	38821 non-null	int64

dtypes: float64(3), int64(7), object(6)

memory usage: 5.0+ MB

None

```

[ ]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

total_reviews = pd.read_csv('DA_airbnb_data_cleaned.csv').
    .groupby('neighbourhood_group')['number_of_reviews'].sum().reset_index()

```

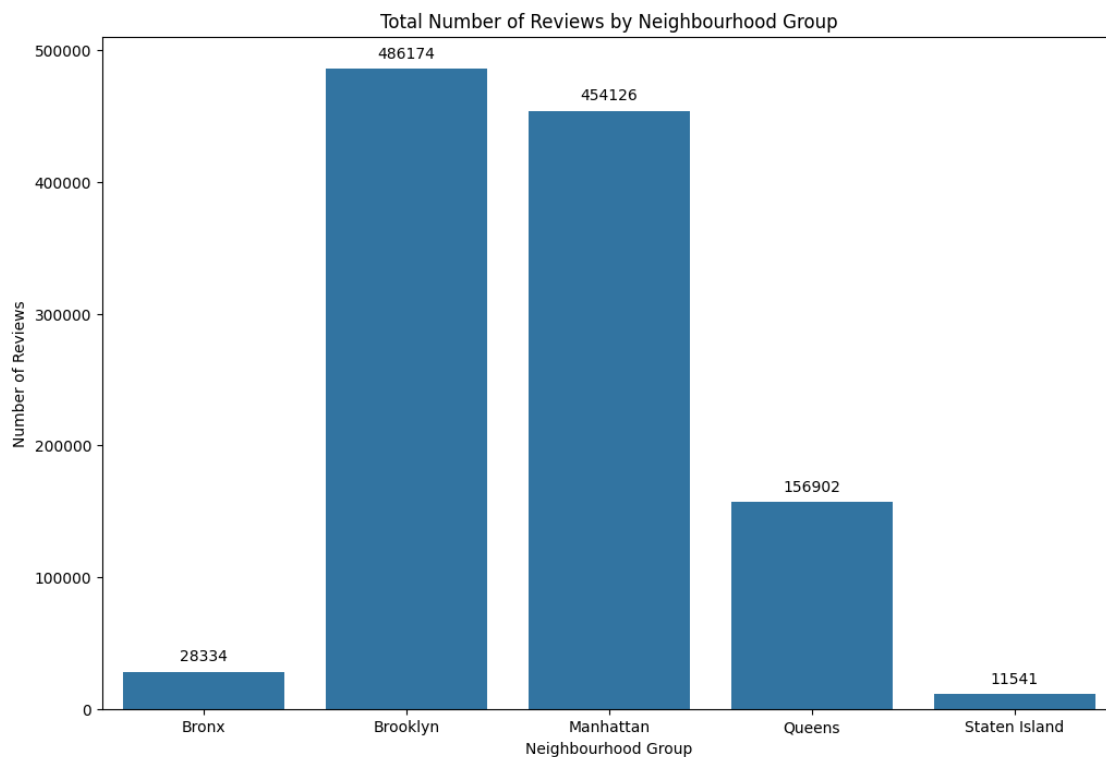
```

plt.figure(figsize=(12, 8))
ax = sns.barplot(x='neighbourhood_group', y='number_of_reviews',
    ↳data=total_reviews)

# Display numbers on top of the bars
for p in ax.patches:
    ax.annotate(f'{int(p.get_height())}', (p.get_x() + p.get_width() / 2., p.
    ↳get_height()),
                ha='center', va='center', xytext=(0, 10), textcoords='offset_
    ↳points')

plt.title('Total Number of Reviews by Neighbourhood Group')
plt.xlabel('Neighbourhood Group')
plt.ylabel('Number of Reviews')
plt.show()

```



```

[ ]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

total_reviews =pd.read_csv('DA_airbnb_data_cleaned.csv').
    ↳groupby('room_type')['number_of_reviews'].sum().reset_index()

```

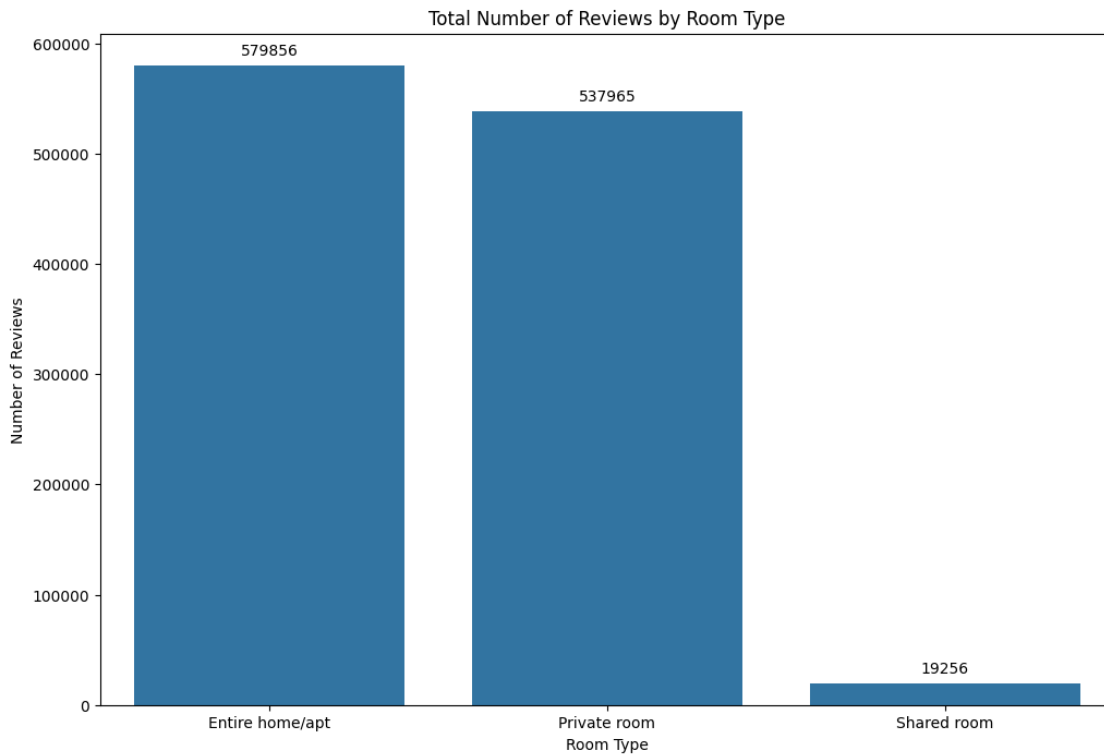
```

plt.figure(figsize=(12, 8))
ax = sns.barplot(x='room_type', y='number_of_reviews', data=total_reviews)

for p in ax.patches:
    ax.annotate(f'{int(p.get_height())}', (p.get_x() + p.get_width() / 2., p.
    ↪get_height()),
                ha='center', va='center', xytext=(0, 10), textcoords='offset_
    ↪points')

plt.title('Total Number of Reviews by Room Type')
plt.xlabel('Room Type')
plt.ylabel('Number of Reviews')
plt.show()

```



```

[ ]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

neighborhood_room_counts = pd.read_csv('DA_airbnb_data_cleaned.csv').
    ↪groupby(['neighbourhood_group', 'room_type']).size().
    ↪reset_index(name='count')

```

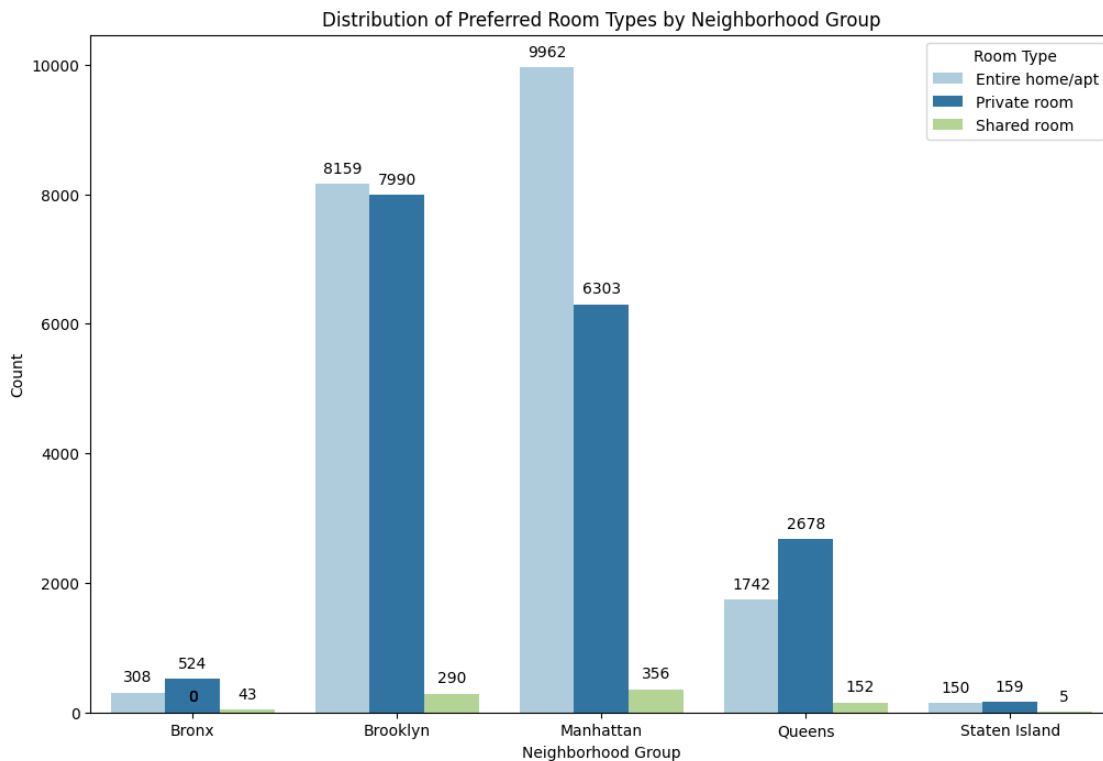
```

plt.figure(figsize=(12, 8))
ax = sns.barplot(x='neighbourhood_group', y='count', hue='room_type',
    ↪data=neighborhood_room_counts, palette='Paired')

for p in ax.patches:
    ax.annotate(f'{{int(p.get_height())}}', (p.get_x() + p.get_width() / 2., p.
    ↪get_height()),
    ↪ha='center', va='center', xytext=(0, 10), textcoords='offset_
    ↪points')

plt.title('Distribution of Preferred Room Types by Neighborhood Group')
plt.xlabel('Neighborhood Group')
plt.ylabel('Count')
plt.legend(title='Room Type', bbox_to_anchor=(1, 1))
plt.show()

```



```

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```

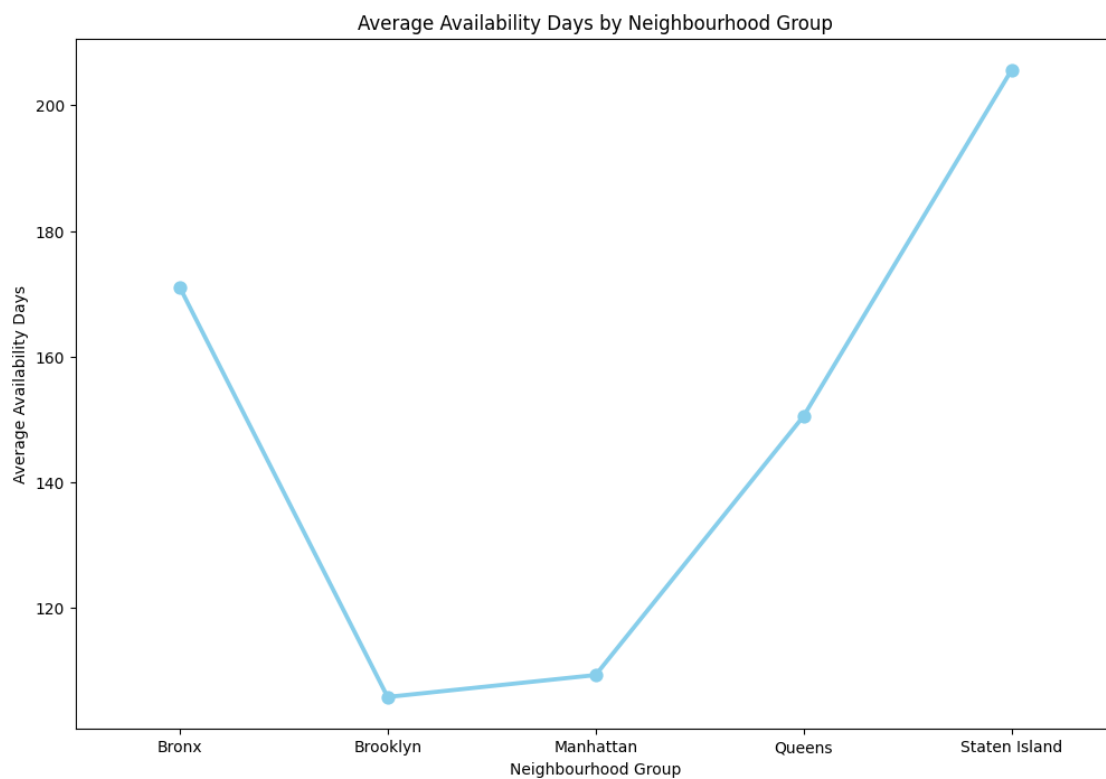
```

# To Calculate the average availability_365 per neighborhood group
avg_availability = pd.read_csv('DA_airbnb_data_cleaned.csv').
    ↳groupby('neighbourhood_group')['availability_365'].mean().reset_index()

plt.figure(figsize=(12, 8))
sns.pointplot(x='neighbourhood_group', y='availability_365',
    ↳data=avg_availability, color='skyblue')

plt.title('Average Availability Days by Neighbourhood Group')
plt.xlabel('Neighbourhood Group')
plt.ylabel('Average Availability Days')
plt.show()

```



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